# Atropine Treatment for Hypertrophic Pyloric Stenosis: A Systematic Review and Meta-analysis

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## Abstract

**Introduction** Several authors have reported the use of atropine as an alternative treatment to pyloromyotomy in infants with hypertrophic pyloric stenosis (HPS). Our aims were to review the efficacy of atropine in treating HPS and to compare atropine therapy versus pyloromyotomy.

**Materials and Methods** Using a defined search strategy (PubMed, MEDLINE, OVID, Embase, Cochrane databases), two investigators independently identified studies reporting the use of atropine for HPS. Case reports and opinion articles were excluded. Outcome measures included success rate, side effects, and length of hospital stay. Maneuvers were compared using Fisher's exact test, and meta-analysis was conducted using RevMan 5.3. Data are expressed as mean ± standard deviation.

**Results** *Systematic review:* of 2,524 abstracts screened, 51 full-text articles were analyzed. There were no prospective or randomized studies. Twelve articles (508 infants) reported HPS resolution using atropine in 402 (79.1%) patients. Atropine side effects were documented in 38/251 (15.1%) infants and included tachycardia, increased transaminases, and flushed skin. *Meta-analysis:* five studies compared atropine treatment (293 infants) with pyloromyotomy (537 infants). Pyloromyotomy had higher success rate (100%) than atropine (80.8%; $p < 0.01$) and shorter hospital stay ($5.6 \pm 2.3$ vs. $10.3 \pm 3.8$ days, respectively; $p < 0.0001$).

**Conclusions** Comparative but nonrandomized studies indicate that atropine is less effective than pyloromyotomy to treat infants with HPS. Currently, there is no evidence-based literature to support atropine treatment in these infants. To our knowledge, atropine should be reserved for patients unfit for general anesthesia or surgery.

**Keywords**
- atropine
- pyloromyotomy
- hypertrophic pyloric stenosis
- systematic review
- meta-analysis



## Introduction

Hypertrophic pyloric stenosis (HPS) is a condition characterized by hypertrophy of the circular muscle of the pylorus, causing pyloric channel narrowing and elongation.[1] Standard treatment for HPS is surgical and involves an extramucosal pyloromyotomy as described by Ramstedt in 1912.[1,2] This procedure has remained unmodified for over a century, except for the renowned minimal access modifications that include the circumumbilical incision described by Tan and Bianchi in 1986,[3] and the laparoscopic approach described by Alain et al in 1990.[4] The laparoscopic approach has become popular over the last two decades, and according to a randomized controlled trial, it is associated with shortened time to full feed and reduced postoperative hospital stay, compared with the open procedure.[5] Moreover, a systematic

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review and meta-analysis on open versus laparoscopic pyloromyotomy for HPS has shown that the postoperative complication rate is similar between the two techniques.[6]

Although surgery for HPS is safe and effective, some authors have investigated less invasive forms of treatment for this condition. During the 1950s and 1960s, antimuscarinic agents such as atropine[7] and scopolamine[8–10] were tested in infants with HPS but reached unsatisfactory success rate. Given the good results obtained with pyloromyotomy in patients with HPS,[11] nonsurgical treatment options were abandoned, and no study on medical treatment of HPS was reported in the literature for decades. However, in 1996, a group of surgeons from Osaka, Japan, published promising results on the use of atropine for infants with HPS.[12] Interestingly, this study was followed by other reports that described the positive effects of atropine in these patients.[13–23] Atropine is an anticholinergic agent with strong antimuscarinic effects that induce smooth muscle relaxation. The rationale for its use in patients with HPS is to induce relaxation of the contracted hypertrophic musculature of the pylorus and reestablish normal feed passage through it.

As in the past years the use of atropine in infants with HPS has been tested in few studies, we have investigated the literature to review the efficacy of this medication in treating infants with HPS and to compare the success and complication rates of atropine treatment with those of pyloromyotomy.

## Materials and Methods

Both the systematic review and the meta-analysis were drafted with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.[24] Two different health librarians were involved: the Bibl@Ud'A ("d'Annunzio" University of Chieti-Pescara, Italy) and the Gerstein Science Information Centre (University of Toronto, Ontario, Canada).

### Systematic Review

This study was registered on PROSPERO, an international prospective register of systematic reviews (registration number: CRD42016051418).[25]

A systematic review of the literature was made using a defined search strategy. Two investigators (G. L. and V. C.) independently searched scientific databases (PubMed, Medline, Cochrane Collaboration, Embase, and Web of Science) using a combination of keywords (►Table 1). MeSH headings and terms used are "pyloric stenosis AND atropine" and "pyloric stenosis AND treatment" (►Supplementary Table S1, online-only). Case reports, opinion articles, and case series with less than 10 patients were excluded.

All gray literature publications (i.e., reports, theses, conference proceedings, bibliographies, commercial documentations, and official documents not published commercially) were excluded.

The full text of the potentially eligible studies was retrieved and independently assessed for eligibility by the same two investigators. Any disagreement between them over the eligibility of particular studies was resolved through discussion with a third author (A.Z.). Outcome measures included atropine success rate and side effects.

**Table 1** Inclusion criteria of systematic review

| Publication | |
|---|---|
| Language | Any |
| Date | After 1950 |
| Subject | Human studies |
| Study type | Retrospective |
| | Prospective |
| | Case–control |
| | Cohort |
| Excluded | Case reports |
| | Case series |
| | Letters |
| | Editorials |
| | Gray literature |
| Keywords | pyloric stenosis |
| | hypertrophic pyloric stenosis |
| | infantile hypertrophic pyloric stenosis |
| | atropine |
| | treatment |

### Meta-analysis

Only studies comparing atropine versus pyloromyotomy for the treatment of HPS were included. Outcome measures included success rate, treatment complications, and length of hospital stay.

Meta-analysis was conducted with RevMan 5.3,[26] using the random-effects model to produce risk ratio (RR) for categorical variables and mean differences (MDs) for continuous variables, along with 95% confidence intervals (CI). We produced $I^2$ values to assess homogeneity. Publication biases were assessed using the funnel plot method.

Atropine treatment and pyloromyotomy were compared with regard to success rate using Fisher's exact test. Data are expressed as mean $\pm$ standard deviation.

### Quality Assessment

Two investigators (G.L. and V.C.) independently assessed the quality and then came to a consensus of all papers that met our inclusion criteria using the Cochrane "risk of bias" tool for comparative studies.[27] Two senior authors (P.L.C. and A.P.) independently evaluated this systematic reviews and meta-analysis using *a measurement tool to assess systematic reviews* (AMSTAR).[28] The PRISMA checklist of our study was then completed.[24]

## Results

### Systematic Review

Of the 2,524 titles screened, 1,612 abstracts were analyzed, and 51 full-text articles were examined. Of these, 12 papers (508 infants) met our inclusion criteria (►Fig. 1 and ►Table 2).[12–23] None of these articles was a prospective or randomized study.

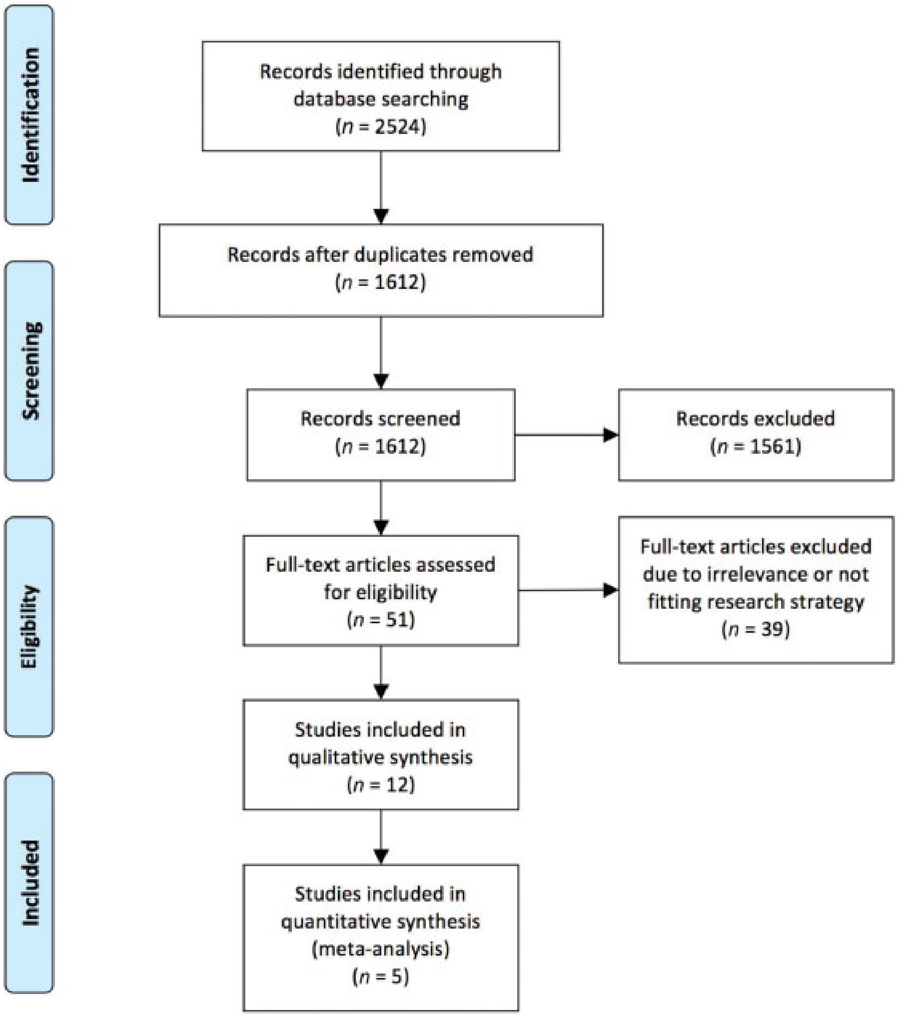


**Fig. 1** Diagram of workflow in the systematic review and meta-analysis.

Atropine was successful in treating HPS in 402 (79.1%) of the 508 patients.[12–23] Atropine side effects were documented in 38 (15.1%) of 251 infants and included tachycardia, increased transaminases, flushed skin, urinary tract infection, upper respiratory tract infection, and hematemesis (►Table 2).[12,13,15,17–19,21,23]

### Meta-analysis

Only five studies compared atropine therapy for HPS (293 infants) with pyloromyotomy (537 infants; ►Table 3).[13,18,21–23] Pyloromyotomy showed a higher success rate (100%) than atropine (80.8%; RR 17.32 [CI: 4.45–67.52]; $p < 0.0001$; ►Fig. 2). Furthermore, there was significant homogeneity between the studies ($I^2 = 12\%$; $p =$ ns), and the funnel plot of published studies demonstrated a convincing symmetry indicating no potential publication bias, although symmetry is difficult to determine with only five studies contributing to the funnel plot (►Fig. 3).

Incidence of complications was similar between the patients treated with atropine (10/118; 8.5%) and those who underwent pyloromyotomy (8/123, 6.5%; RR 1.11 [CI 0.16–7.85]; $p =$ ns; ►Fig. 4), even if there was a relevant albeit not significant heterogeneity between the studies ($I^2 = 62\%$; $p = 0.07$). Atropine therapy caused only minor complications (►Table 2).[13,18,23] Six out of eight complications related to pyloromyotomy were minor ones (all wound infections), and two were majors: one bowel mucosal perforation and one postoperative hemorrhagic shock in a patient with hemophilia.[13,18]

Moreover, patients treated with pyloromyotomy had a shorter length of hospital stay (5.6 ± 2.3 days) than those treated with atropine (10.3 ± 3.8 days; MD 4.52 [CI 2.43–6.60]; $p < 0.0001$; ►Fig. 5). However, there was significant heterogeneity between the studies ($I^2 = 82\%$; $p < 0.001$).

## Discussion

This systematic review and meta-analysis shows that pyloromyotomy has a higher success rate than atropine treatment (either oral or intravenous [IV]) in infants with HPS. In fact, pyloromyotomy reaches a 100% success rate, whereas atropine's overall success rate is approximately 80%. Moreover, atropine treatment for HPS is associated with an increased length of hospital stay.

**Table 2** Studies reporting atropine treatment in infantile hypertrophic pyloric stenosis

| Author | Year | Type of study | Atropine doses (via) | Atropine, *n* | Success, *n* (%) | Collateral effects, *n* (%) | Collateral effects description |
|---|---|---|---|---|---|---|---|
| Nagita et al[12] | 1996 | R | 0.04–0.11 mg/kg/d (IV) | 22 | 21 (95.5) | 6 (27.3) | Flushed skin (3), increased AST/ALT (2), tachycardia (1) |
| Yamataka et al[13] | 2000 | R | 0.05–0.1 mg/kg/d (orally, then IV) | 14 | 12 (85) | 1 (7.1) | Hematemesis (1) |
| Riccabona et al[14] | 2001 | R | 1 drop/6–8 times per day (orally) | 22 | 7 (32) | n.r. | |
| Singh et al[15] | 2001 | R | 0.06–0.255 mg/kg/d (IV), 0.12–0.66 mg/kg/d (orally) | 52 | 50 (96.2) | 11 (21.2) | Increased AST/ALT (8), tachycardia (3) |
| Sretenović et al[16] | 2004 | R | 0.05–0.1 mg/kg/d (orally) | 22 | 18 (81.8) | n.r. | |
| Singh et al[17] | 2005 | R | 0.18 mg/kg/d (orally) | 12 | 11 (91.6) | 0 (0) | |
| Kawahara et al[18] | 2005 | R | 0.06 mg/kg/d (IV), then 0.12 mg/kg/d (orally) | 52 | 45 (87) | 3 (5.8) | UTI (1), increased AST/ALT (1), upper respiratory tract infection (1) |
| Meissner et al[19] | 2006 | R | 0.04–0.11 mg/kg/d (IV), then doubled (orally) | 33 | 25 (75.8) | 11 (33.3) | Transient tachycardia (8), flushed skin (2), UTI (1) |
| Koike et al[20] | 2013 | R | 0.04–0.1 mg/kg/d (IV), then doubled (orally) | 33 | 18 (58) | n.r. | |
| Lukac et al[21] | 2013 | R | 0.05–0.18 mg/kg/d (orally) | 40 | 30 (75) | 0 (0) | |
| Takeuchi et al[22] | 2013 | ND | n.r. | 180 | 142 (78.9) | n.r. | |
| Fan et al[23] | 2016 | P | 0.06 mg/kg/d (step-up IV) then step-down (orally) | 26 | 23 (88.5) | 6 (23) | Transient tachycardia and flushed skin (6) |

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; IV, intravenously; n.r., not reported; ND, nationwide database; P, prospective not randomized study; R, retrospective study; UTI, urinary tract infection.

**Table 3** Studies comparing atropine versus surgery (pyloromyotomy) in infantile hypertrophic pyloric stenosis

| | | | | | Success, *n* (%) | | Complications, *n* (%) | | Hospital stay mean (range) | |
|---|---|---|---|---|---|---|---|---|---|---|
| Author | Year | Type of study | Atropine, *n* | Surgery, *n* | Atropine | Surgery | Atropine | Surgery | Atropine | Surgery |
| Yamataka et al[13] | 2000 | R | 14 | 20 | 12 (85) | 20 (100) | 1 (7.1) | 2 (10) | 5.3 (1–10) | 2.7 (2–3) |
| Kawahara et al[18] | 2005 | R | 52 | 40 | 45 (87) | 40 (100) | 3 (5.8) | 6 (15) | 13 (6–36) | 5 (4–29) |
| Lukac et al[21] | 2013 | R | 40 | 26 | 30 (75) | 26 (100) | 0 (0) | 0 (0) | n.r. | n.r. |
| Takeuchi et al[22] | 2013 | ND | 180 | 435 | 142 (78.9) | 435 (100) | n.r. | n.r. | 13.5 (n.r.) | 8.0 (n.r.) |
| Fan et al[23] | 2016 | P | 26 | 23 | 23 (88.5) | 26 (100) | 6 (23) | 0 (0) | 9.5 (6–11) | 6.8 (5–9) |

Abbreviations: n.r., not reported; ND, nationwide database; P, prospective not randomized study; R, retrospective study.

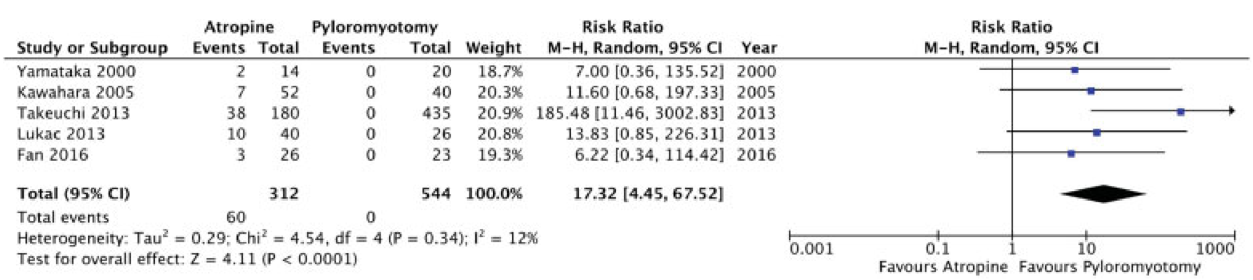


**Fig. 2** Forest plot comparison of success rate in infants with hypertrophic pyloric stenosis who were treated with atropine versus pyloromyotomy. Events indicate patients with persistency of hypertrophic pyloric stenosis.

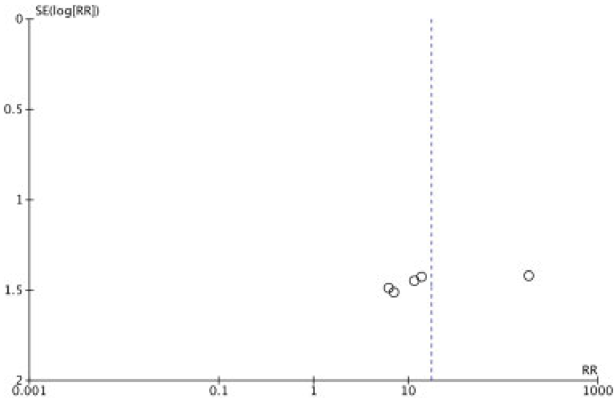


**Fig. 3** Funnel plot of included studies comparing the success rate in infants with hypertrophic pyloric stenosis.

The etiology of HPS is still not clear.[29–31] Kawahara et al demonstrated characteristic abnormal contractile activity in the gastroduodenal junction of infants affected by HPS.[32] The gastroduodenal junction is uncoordinated with the antral contraction, hence presumably resulting in spasm of the pylorus muscle and subsequently in hypertrophy. Atropine is a cholinergic blocking agent with known peripheral antimuscarinic activity, leading to decreased gastrointestinal peristalsis by smooth muscle relaxation.[19,33] Because of this, atropine has been used as an antispasmodic agent for gastrointestinal disorders.[12] For the same reason, antimuscarinic agents, such as atropine and scopolamine, were used in the 1950s and 1960s as medical management of infants with HPS. Jacoby reported in 1962 that oral atropine methyl nitrate was successful in approximately 90% of children with HPS. However, in the same study, the success rate of pyloromyotomy was approximately 97%.[7] Similarly, two other studies showed that scopolamine successfully treated infants with HPS in 60 to 70% of cases.[9,10] On the contrary, in 1968, Schärli et al reported a success rate of pyloromyotomy close to 99% in patients with HPS treated from 1961 to 1967, with a mortality decreased to 0.5% over the previous decades.[11] Therefore, either because of the unsatisfactory outcome of the medical management or because of the improving results of pyloromyotomy over the same period of time, at the end of the 1960s, antimuscarinic agents were abandoned in favor of the more efficient surgical procedure for the treatment of HPS.[22]

Two decades ago, Nagita et al reported promising results with atropine treatment for HPS (success rate in 21/22 infants; 95.5%), thus resulting in a renewed interest in this medical management.[12] Authors have highlighted the importance of the dosage and the way of administration of atropine. As known, the pharmacologic activity of IV atropine is two to three times greater than that of the oral (PO) form, with faster response to the effective IV dose.[30] To achieve effective blood concentration, atropine has to be given either intravenously or orally at a dose twice as high as the effective IV dose. However, both the IV form and the high-dose PO form may be associated with more adverse effects because the half-life elimination of atropine in children aged <2 years takes two to three times longer than in children aged >2 years.[12] Nonetheless, cardiac side effects of atropine in normal hearts are rare and are generally limited to sinus tachycardia.[34] Noncardiac adverse effects in healthy children include dilated pupils, dry hands and mouth, and pro-hyperthermia. Thus, for children with critical care illnesses the pro-hyperthermic effects of atropine should not be underestimated.[35]

This systematic review showed that the success rate of atropine (79.1%) and the incidence of side effects (15.1%) were similar to those reported by two recent meta-analyses.[29,36] However, atropine success rate was inhomogeneous among the included studies, ranging between 32 and 96%. This inhomogeneity could be explained by the wide variability of atropine effective range, which is linked to the variability of muscarinic receptor sensitivity in the muscle,

| | Atropine | | Pyloromyotomy | | | Risk Ratio | |
|---|---|---|---|---|---|---|---|
| Study or Subgroup | Events | Total | Events | Total | Weight | M-H, Random, 95% CI | Year |
| Yamataka 2000 | 1 | 14 | 2 | 20 | 31.0% | 0.71 [0.07, 7.13] | 2000 |
| Kawahara 2005 | 3 | 52 | 6 | 40 | 43.5% | 0.38 [0.10, 1.44] | 2005 |
| Lukac 2013 | 0 | 26 | 0 | 40 | | Not estimable | 2013 |
| Fan 2016 | 6 | 26 | 0 | 23 | 25.5% | 11.56 [0.69, 194.53] | 2016 |
| **Total (95% CI)** | | **118** | | **123** | **100.0%** | **1.11 [0.16, 7.85]** | |
| Total events | 10 | | 8 | | | | |

Heterogeneity: Tau² = 1.84; Chi² = 5.24, df = 2 (P = 0.07); I² = 62%
Test for overall effect: Z = 0.10 (P = 0.92)

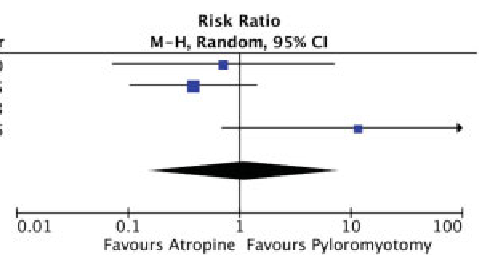


**Fig. 4** Forest plot comparison of complications incidence in infants with hypertrophic pyloric stenosis who were treated with atropine versus pyloromyotomy.

| | Atropine | | | Pyloromyotomy | | | | Mean Difference | |
|---|---|---|---|---|---|---|---|---|---|
| Study or Subgroup | Mean | SD | Total | Mean | SD | Total | Weight | IV, Random, 95% CI | Year |
| Yamataka 2000 | 5.3 | 2.9 | 14 | 2.7 | 0.5 | 20 | 26.3% | 2.60 [1.07, 4.13] | 2000 |
| Kawahara 2005 | 13 | 7.5 | 52 | 5 | 6.25 | 40 | 19.7% | 8.00 [5.19, 10.81] | 2005 |
| Takeuchi 2013 | 13.5 | 6.7 | 180 | 8 | 2.9 | 435 | 28.6% | 5.50 [4.48, 6.52] | 2013 |
| Fan 2016 | 9.5 | 3.7 | 26 | 6.8 | 2.4 | 23 | 25.4% | 2.70 [0.97, 4.43] | 2016 |
| **Total (95% CI)** | | | **272** | | | **518** | **100.0%** | **4.52 [2.43, 6.60]** | |

Heterogeneity: Tau² = 3.68; Chi² = 19.73, df = 3 (P = 0.0002); I² = 85%
Test for overall effect: Z = 4.25 (P < 0.0001)

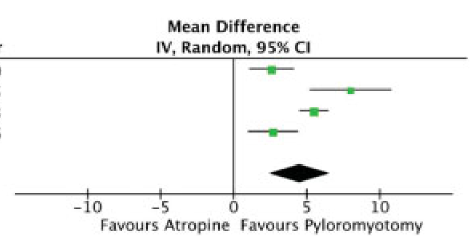


**Fig. 5** Forest plot comparison of length of hospital stay in infants with hypertrophic pyloric stenosis who were treated with atropine versus pyloromyotomy.

drug clearance, blood flow, and pyloric innervation.[29] Moreover, the inhomogeneity reflects the significant discrepancies between the protocols adopted at different centers, especially for atropine administration route and its dosage. Some authors adopted a step-up approach, that is, increasing the dose of atropine, or switching from PO to IV in unsuccessful cases or, a step-down approach, from IV to PO in infants with positive response.[12,13,15,16,18–21,23] Of note, the maximum dose either PO or IV could vary in these stepwise approaches. The most promising results obtained with atropine therapy were achieved by Singh et al,[15] thanks to the highest doses of the drug given either intravenously or orally (96.2% success rate; ►Table 2). However, as a consequence, authors reported a relevant rate of collateral effects (21.2%).

To the best of our knowledge, this is the first meta-analysis including studies comparing atropine versus pyloromyotomy in infants with HPS, whereas previous reviews included also noncomparative studies.[29,36] However, the risk of bias assessments for the individual comparing studies included in this meta-analysis varied quite a bit (►Supplementary Table S2, online-only).

Based on our meta-analysis, atropine has a success rate of 80%, whereas pyloromyotomy has a success rate of 100%. The forest plot comparison of this primary outcome is based on five studies, with a nonsignificant degree of variability. Essentially, while in the surgical arm, the studies were consistent as all centers performed a pyloromyotomy (with an open or laparoscopic approach), in the medical arm, all protocols differ with regard to atropine doses and administration route. In fact, a study adopted a step-up protocol from PO to IV treatment in unsuccessful infants,[13] another paper adopted a fixed dose of atropine that was first administered intravenously and then orally in patients with no episodes of vomiting for more than 24 hours,[18] other authors applied a mere step-up dose of PO atropine,[21] and the last study used a step-up IV dose followed by a step-down PO treatment.[23]

The incidence of complications was similar for both treatment groups. All the complications caused by atropine therapy were minor side effects. Conversely, two out of six complications in the surgical arm were majors: one mucosal perforation during pyloromyotomy and one postoperative hemorrhagic shock in a patient with hemophilia.[18]

As expected, patients treated with pyloromyotomy had a reduced length of hospital stay than those treated with atropine. This outcome measure was reported in four studies, in which a step-up protocol[13,24] seemed to reach better results than a fixed dose of atropine.[18] However, criteria for hospital stay were arbitrarily reported by centers, thus partially explaining the relevant length of hospital stay in infants treated with pyloromyotomy.

## Limitations

We acknowledge the limitations of our study, which, as any other meta-analysis, relies on the quality of the studies and data available in the literature. Some of the limitations of our current meta-analysis are due to the retrospective nature of studies as well as due to the variability in atropine protocols, including drug administration route and dosage. Statistical heterogeneity of data was fairly low for all of our analyses, with the exception of the analysis of complications. Despite this, clinical heterogeneity does exist and is inherent in meta-analysis, and this must be considered when interpreting the results of the study. However, to the best of our knowledge, this paper is the only one in the literature that attempted to validate an evidence-based approach for medical management of HPS. Furthermore, when independently assessed by two senior authors using AMSTAR, this systematic review and meta-analysis received a relevant score (►Supplementary Table S3, online-only). Finally, the PRISMA checklist of our study was completed (►Supplementary Table S4, online-only).

## Conclusion

In conclusion, there is no current evidence-based support for the use of atropine in infants with HPS. Comparative but nonrandomized studies indicate that atropine is less effective than pyloromyotomy in the treatment of HPS. Although the incidence of complications is comparable between the two groups, pyloromyotomy significantly shortens the length of hospital stay. Therefore, we recommend the use of atropine only in selected infants with HPS, such as those with a hostile abdomen (e.g., infants with a history of major abdominal surgical procedure, such as giant exomphalos or necrotizing enterocolitis) or those at high risk for general anesthesia or surgical procedures (e.g., hemophilia).

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